PROBLEM 39

# Particle on a String over a Plate Edge

Original statement · Problems.pdf, p. 27

String + Edge Constraint

A heavy point particle $P$ with mass $m$ is connected with point $O$ on the upper surface of a thin square plate by a massless, inextensible string of length $\ell$. The distance between $O$ and the edge of the plate is $d$. The string hangs over the edge of the plate and can slide without friction. You can assume that the string remains tense and in contact with the plate at all times.

We denote the geometric point where the string is in contact with the plate edge as $Q$. We define the inertial reference frame $S_0 : \{O, \mathcal{B}_0\}$, with $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ as shown in the figure (Problems.pdf, PDF page 27). We define the angle $\theta$ between the $Ox_0$ axis and the string segment $OQ$, and the angle $\phi$ between a vertical plane containing the plate edge and the line perpendicular to the plate edge that passes through $P$. Initially at time $t = 0, \theta(0) = \phi(0) = \pi/3$ rad, and the modulus of the velocity of $P$ is $v_0^P(0) = \sqrt{\ell g}$, where $g$ is the magnitude of the acceleration due to gravity.

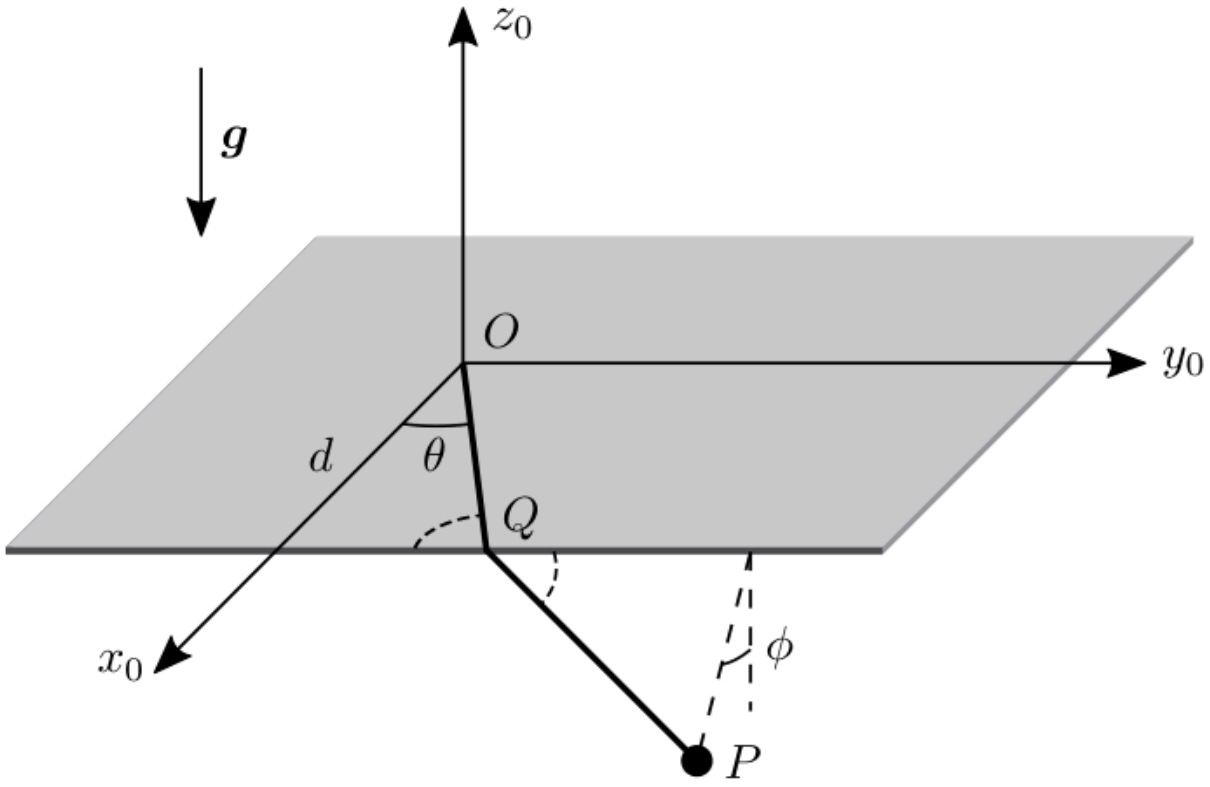



*Figure: Plate with O on its upper surface, string OQ at angle θ to Ox0, hanging segment QP at angle φ, equal angles marked with dashed lines (Problems.pdf, p. 27).*

*Hint: Note that under the conditions of the problem, the two angles marked with dashed lines are equal.*

**(a)** How many degrees of freedom does the system have?

**(b)** Obtain the general expression of a unit vector in the direction from $Q$ to $P$ as a function of $\theta$ and $\phi$.

**(c)** Obtain the general expression for the position, velocity, and acceleration vectors of $Q$ and $P$ in $S_0$ as a function of $\theta$, $\phi$, their derivatives, and the parameters of the problem.

**(d)** Identify and express mathematically all the forces acting on $P$.

**(e)** Find a set of second order differential equations to determine the evolution of $\theta$ and $\phi$ in time, free of any other unknowns.

**(f)** Does the tension force exert work on $P$? Provide proof of your answer.

**(g)** Derive the general expression of the mechanical energy of $P$ in $S_0$ and compute its value initially. Is this a conserved quantity of motion? Justify your answer.

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **System:** heavy particle $P$ on a massless string of length $\ell$ that slides without friction over the straight edge of a plate; $Q$ is the contact point.
- **Hypotheses:** $S_0$ is inertial and $\mathbf{g} = -g\mathbf{k}_0$ ($Oz_0$ up). The edge is the line $x_0 = d$, $z_0 = 0$, parallel to $\mathbf{j}_0$ (read from the figure). The edge is smooth and the string is massless, so the tension $T \geq 0$ is the same on both sides of $Q$ and the edge force on the string is normal to the edge. $P$ does not touch the plate. No drag.
- **Degrees of freedom:** $P$ has 3 coordinates and one holonomic constraint $\|\mathbf{OQ}\| + \|\mathbf{QP}\| = \ell$, so $\mathrm{CDOF} = 3 - 1 = 2$ with generalized coordinates $(\theta, \phi)$.
- **Initial state:** $\theta(0) = \phi(0) = \pi/3$ and $\|\mathbf{v}_0^P(0)\| = \sqrt{\ell g}$. The statement gives only the speed, not its direction; for numerical checks $\dot{\phi}(0) = 0$ is assumed (the initial energy does not depend on it).

| Symbol | Meaning | SI unit | Value in the numerical check |
|---|---|---|---|
| $m$ | mass of $P$ | kg | 1 |
| $\ell$ | string length | m | 2.0 |
| $d$ | distance from $O$ to the edge | m | 0.4 |
| $g$ | gravitational acceleration | m/s$^2$ | 9.81 |

At $t = 0$ the hanging length is $\ell - d/\cos(\pi/3) = \ell - 2d$, so the data must satisfy $\ell > 2d$.

**Assumptions and absence of an official key.** The official key (Problems.pdf, PDF page 76) answers only part (f). Parts (a)-(e) and (g) are derived here from the statement and the figure on PDF page 27, with two explicit assumptions: the edge is the line

$x_0 = d$ parallel to $\mathbf{j}_0$, and $\phi$ is measured as described in the statement. For those parts there is no official result to compare with, so no discrepancy can be reported; the result for (f) agrees with the key.

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- **Contact point $Q$.** It lies on the edge with $OQ$ at angle $\theta$ from $Ox_0$:

$$\mathbf{r}_0^{Q} = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0, \qquad \|\mathbf{OQ}\| = \frac{d}{\cos\theta}, \qquad \hat{\mathbf{u}}_{OQ} = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0$$

**Hanging length:** $\rho \equiv \|\mathbf{QP}\| = \ell - d/\cos\theta$. Define $w \equiv \rho\cos\theta = \ell\cos\theta - d > 0$.

**Why the dashed angles are equal.** The string element at $Q$ is massless, so the forces on it add to zero: $-T\hat{\mathbf{u}}_{OQ}$ (pull towards $O$), $+T\hat{\mathbf{u}}_{QP}$ (pull towards $P$) and the edge force $\mathbf{N}$. A smooth edge pushes only normally to itself, so the $\mathbf{j}_0$-component gives $-T\,\hat{\mathbf{u}}_{OQ}\cdot\mathbf{j}_0 + T\,\hat{\mathbf{u}}_{QP}\cdot\mathbf{j}_0 = 0$, that is $\hat{\mathbf{u}}_{QP}\cdot\mathbf{j}_0 = \sin\theta$, the statement of the hint.

**Part (b).** The $\mathbf{j}_0$-component of $\hat{\mathbf{u}}_{QP}$ is $\sin\theta$; the remainder has modulus $\sqrt{1-\sin^2\theta} = \cos\theta$ and lies in the plane normal to the edge, where a downward line making the angle $\phi$ with the vertical plane $x_0 = d$ has direction $\sin\phi\,\mathbf{i}_0 - \cos\phi\,\mathbf{k}_0$. Hence

$$\hat{\mathbf{u}} \equiv \hat{\mathbf{u}}_{QP} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0, \qquad \|\hat{\mathbf{u}}\|^2 = \cos^2\theta\left(\sin^2\phi + \cos^2\phi\right) + \sin^2\theta = 1$$

**Moving basis** $\mathcal{B}_1 = \{\hat{\mathbf{u}}, \mathbf{e}_\theta, \mathbf{e}_\phi\}$. Differentiating $\hat{\mathbf{u}}$ with respect to $\theta$ and $\phi$ gives $\partial\hat{\mathbf{u}}/\partial\theta = \mathbf{e}_\theta$ and $\partial\hat{\mathbf{u}}/\partial\phi = \cos\theta\,\mathbf{e}_\phi$ with the unit vectors

$$\mathbf{e}_\theta = -\sin\theta\sin\phi\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0 + \sin\theta\cos\phi\,\mathbf{k}_0, \qquad \mathbf{e}_\phi = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0$$

Change-of-basis matrix (columns are the components of $\hat{\mathbf{u}}, \mathbf{e}_\theta, \mathbf{e}_\phi$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta\sin\phi & -\sin\theta\sin\phi & \cos\phi \\ \sin\theta & \cos\theta & 0 \\ -\cos\theta\cos\phi & \sin\theta\cos\phi & \sin\phi \end{pmatrix}$$

- **Orthonormality ($R^TR = I$):** $\hat{\mathbf{u}}\cdot\hat{\mathbf{u}} = 1$; $\mathbf{e}_\theta\cdot\mathbf{e}_\theta = \sin^2\theta\sin^2\phi + \cos^2\theta + \sin^2\theta\cos^2\phi = 1$; $\mathbf{e}_\phi\cdot\mathbf{e}_\phi = 1$; $\hat{\mathbf{u}}\cdot\mathbf{e}_\theta = \sin\theta\cos\theta\left(1 - \sin^2\phi - \cos^2\phi\right) = 0$; $\hat{\mathbf{u}}\cdot\mathbf{e}_\phi = \cos\theta\sin\phi\cos\phi - \cos\theta\cos\phi\sin\phi = 0$; $\mathbf{e}_\theta\cdot\mathbf{e}_\phi = -\sin\theta\sin\phi\cos\phi + \sin\theta\cos\phi\sin\phi = 0$.
- **Determinant:** component by component, $\hat{\mathbf{u}}\times\mathbf{e}_\theta = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0 = \mathbf{e}_\phi$ (using $\sin^2\theta + \cos^2\theta = 1$), so the triad is right-handed and $\det[{}_0R_1] = +1$.

**Derivatives of the moving basis** (from the partial derivatives, $\dot{\mathbf{e}}_i = \dot\theta\,\partial\mathbf{e}_i/\partial\theta + \dot\phi\,\partial\mathbf{e}_i/\partial\phi$):

$$\dot{\hat{\mathbf{u}}} = \dot\theta\,\mathbf{e}_\theta + \dot\phi\cos\theta\,\mathbf{e}_\phi, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,\hat{\mathbf{u}} - \dot\phi\sin\theta\,\mathbf{e}_\phi, \qquad \dot{\mathbf{e}}_\phi = \dot\phi\,(-\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta)$$

For instance $\partial\mathbf{e}_\theta/\partial\theta = (-\cos\theta\sin\phi, -\sin\theta, \cos\theta\cos\phi) = -\hat{\mathbf{u}}$ and $\partial\mathbf{e}_\phi/\partial\phi = (-\sin\phi, 0, \cos\phi) = -\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta$. The matrix $[{}_0R_1]^T[{}_0\dot R_1]$ built from these relations is skew-symmetric, as required for a rotating orthonormal basis.

PHASE 3

## Step-by-Step Mathematical Deduction

**(a) Degrees of freedom**

Two: $(\theta, \phi)$, as counted in Phase 1.

**(b) Unit vector from $Q$ to $P$**

Obtained in Phase 2: $\hat{\mathbf{u}} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0$.

**(c) Position, velocity and acceleration of $Q$ and $P$**

> **Pedagogical Justification:** $Q$ slides along a straight edge, so only the chain rule on $\tan\theta$ is needed. $P$ is $Q$ plus a vector of variable length along a variable direction, so the product rule is needed.

**Point $Q$** (chain rule $\frac{d}{dt}\tan\theta = \dot\theta/\cos^2\theta$ and $\frac{d}{dt}\cos^{-2}\theta = 2\sin\theta\,\dot\theta/\cos^3\theta$):

$$\mathbf{r}_0^Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0, \qquad \mathbf{v}_0^Q = \frac{d\,\dot\theta}{\cos^2\theta}\,\mathbf{j}_0, \qquad \mathbf{a}_0^Q = \frac{d}{\cos^2\theta}\left(\ddot\theta + 2\tan\theta\,\dot\theta^2\right)\mathbf{j}_0$$

**Position of $P$:**

$$\mathbf{r}_0^P = \mathbf{r}_0^Q + \rho\,\hat{\mathbf{u}}, \qquad \rho = \ell - \frac{d}{\cos\theta}, \qquad \dot\rho = -\frac{d\sin\theta}{\cos^2\theta}\,\dot\theta$$

**Velocity of $P$** (product rule): $\mathbf{v}_0^P = \mathbf{v}_0^Q + \dot\rho\,\hat{\mathbf{u}} + \rho\,\dot{\hat{\mathbf{u}}}$. Project on $\mathcal{B}_1$ with $\mathbf{j}_0\cdot\hat{\mathbf{u}} = \sin\theta, \mathbf{j}_0\cdot\mathbf{e}_\theta = \cos\theta, \mathbf{j}_0\cdot\mathbf{e}_\phi = 0$:

- $\hat{\mathbf{u}}$-component: $\dfrac{d\dot\theta\sin\theta}{\cos^2\theta} + \dot\rho = \dfrac{d\dot\theta\sin\theta}{\cos^2\theta} - \dfrac{d\dot\theta\sin\theta}{\cos^2\theta} = 0.$
- $\mathbf{e}_\theta$-component: $\dfrac{d\dot\theta\cos\theta}{\cos^2\theta} + \rho\dot\theta = \dot\theta\left(\dfrac{d}{\cos\theta} + \ell - \dfrac{d}{\cos\theta}\right) = \ell\dot\theta.$
- $\mathbf{e}_\phi$-component: $0 + \rho\cos\theta\,\dot\phi = w\dot\phi.$

$$\boxed{\mathbf{v}_0^P = \ell\dot\theta\,\mathbf{e}_\theta + w\dot\phi\,\mathbf{e}_\phi}, \qquad \|\mathbf{v}_0^P\|^2 = \ell^2\dot\theta^2 + w^2\dot\phi^2, \qquad w = \ell\cos\theta - d$$

**Acceleration of $P$.** Differentiate with the product rule, using $\dot w = -\ell\sin\theta\,\dot\theta$ and the basis derivatives of Phase 2:

$$\frac{d}{dt}\left(\ell\dot\theta\,\mathbf{e}_\theta\right) = \ell\ddot\theta\,\mathbf{e}_\theta + \ell\dot\theta\left(-\dot\theta\,\hat{\mathbf{u}} - \dot\phi\sin\theta\,\mathbf{e}_\phi\right)$$

$$\frac{d}{dt}\left(w\dot\phi\,\mathbf{e}_\phi\right) = \left(\dot w\dot\phi + w\ddot\phi\right)\mathbf{e}_\phi + w\dot\phi^2\left(-\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta\right)$$

Adding both and collecting terms (the $\mathbf{e}_\phi$ coefficient gathers $-\ell\sin\theta\,\dot\theta\dot\phi$ from the first line and $\dot w\dot\phi = -\ell\sin\theta\,\dot\theta\dot\phi$ from the second):

$$\boxed{\mathbf{a}_0^P = \left(-\ell\dot\theta^2 - w\cos\theta\,\dot\phi^2\right)\hat{\mathbf{u}} + \left(\ell\ddot\theta + w\sin\theta\,\dot\phi^2\right)\mathbf{e}_\theta + \left(w\ddot\phi - 2\ell\sin\theta\,\dot\theta\dot\phi\right)\mathbf{e}_\phi}$$

Components in $\mathcal{B}_0$ follow by multiplying the column of $\mathcal{B}_1$-components by $[{}_0R_1]$.

**(d) Forces on $P$**

> **Pedagogical Justification:** $P$ is a free particle in contact only with the string, so the forces are its weight and the string tension. The edge force acts on the (massless) string at $Q$, not on $P$.

- Weight: $\mathbf{W} = -mg\,\mathbf{k}_0$. With $\mathbf{k}_0\cdot\hat{\mathbf{u}} = -\cos\theta\cos\phi$, $\mathbf{k}_0\cdot\mathbf{e}_\theta = \sin\theta\cos\phi$, $\mathbf{k}_0\cdot\mathbf{e}_\phi = \sin\phi$: $\mathbf{W} = mg\cos\theta\cos\phi\,\hat{\mathbf{u}} - mg\sin\theta\cos\phi\,\mathbf{e}_\theta - mg\sin\phi\,\mathbf{e}_\phi$.
- Tension: $\mathbf{T} = -T\,\hat{\mathbf{u}}$, directed towards $Q$, with $T \geq 0$ unknown.

**(e) Equations of motion free of $T$**

> **Pedagogical Justification:** Newton's second law $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{T}$ has the three unknowns $\ddot\theta, \ddot\phi, T$. Projecting on $\mathcal{B}_1$ isolates $T$ in the $\hat{\mathbf{u}}$ equation, because $\mathbf{T}$ has no component along $\mathbf{e}_\theta$ or $\mathbf{e}_\phi$.

$$\hat{\mathbf{u}}:\ m\left(-\ell\dot\theta^2 - w\cos\theta\,\dot\phi^2\right) = mg\cos\theta\cos\phi - T$$

$$\mathbf{e}_\theta:\ m\left(\ell\ddot\theta + w\sin\theta\,\dot\phi^2\right) = -mg\sin\theta\cos\phi$$

$$\mathbf{e}_\phi:\ m\left(w\ddot\phi - 2\ell\sin\theta\,\dot\theta\dot\phi\right) = -mg\sin\phi$$

The last two contain only $\theta, \phi$ and their derivatives; they are the requested pair (with $w = \ell\cos\theta - d$):

$$\boxed{\ddot{\theta} = -\frac{\sin\theta}{\ell}\left[(\ell\cos\theta - d)\,\dot{\phi}^2 + g\cos\phi\right]}, \qquad \boxed{\ddot{\phi} = \frac{2\ell\sin\theta\,\dot{\theta}\dot{\phi} - g\sin\phi}{\ell\cos\theta - d}}$$

The $\hat{\mathbf{u}}$ equation then gives the tension once the motion is known:

$$T = m\left[g\cos\theta\cos\phi + \ell\dot{\theta}^2 + (\ell\cos\theta - d)\cos\theta\,\dot{\phi}^2\right]$$

which is positive whenever $\cos\phi \geq 0$ and $w > 0$, consistent with a taut string.

**Cross-check by Lagrange's equations.** With kinetic energy $\tfrac{1}{2}m(\ell^2\dot{\theta}^2 + w^2\dot{\phi}^2)$ and $V = mgz_P$, $z_P = \mathbf{r}_0^P \cdot \mathbf{k}_0 = -\rho\cos\theta\cos\phi = -w\cos\phi$, the Lagrangian is $L = \tfrac{1}{2}m(\ell^2\dot{\theta}^2 + w^2\dot{\phi}^2) + mgw\cos\phi$. With $\partial w/\partial\theta = -\ell\sin\theta$:

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}} - \frac{\partial L}{\partial\theta} = m\ell^2\ddot{\theta} + m\ell\sin\theta\left(w\dot{\phi}^2 + g\cos\phi\right) = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\phi}} - \frac{\partial L}{\partial\phi} = m\left(w^2\ddot{\phi} + 2w\dot{w}\dot{\phi}\right) + mgw\sin\phi = 0$$

Dividing the first by $m\ell$ and the second by $mw$ reproduces the two boxed equations. A symbolic comparison at three random states confirmed that the Lagrange expressions equal the $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$ Newton equations multiplied by $\ell$ and $w$ respectively (agreement to machine precision).

**(f) Work of the tension**

> **Pedagogical Justification:** The power of the tension on $P$ is $\mathbf{T}\cdot\mathbf{v}_0^P$; it vanishes if $\mathbf{v}_0^P \perp \hat{\mathbf{u}}$.

$$P_T = \mathbf{T}\cdot\mathbf{v}_0^P = -T\,\hat{\mathbf{u}}\cdot\left(\ell\dot{\theta}\,\mathbf{e}_\theta + w\dot{\phi}\,\mathbf{e}_\phi\right) = 0$$

because $\hat{\mathbf{u}}$ is orthogonal to $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$. Equivalently, the $\hat{\mathbf{u}}$-component of $\mathbf{v}_0^P$ in (c) cancelled: the string slides over the edge so that the hanging length grows at the rate at which $Q$ moves along $\hat{\mathbf{u}}$. **No: the tension does no work on** $P$ (official key, PDF page 76: $\hat{\mathbf{u}} \cdot \mathbf{v}_0^P = 0$).

**(g) Mechanical energy**

**General expression** (kinetic energy from (c); gravitational potential energy with $Oz_0$ up and $V = 0$ on the plate plane):

$$E = \tfrac{1}{2} m \|\mathbf{v}_0^P\|^2 + m g z_P = \tfrac{1}{2} m \left[ \ell^2 \dot{\theta}^2 + (\ell \cos\theta - d)^2 \, \dot{\phi}^2 \right] - m g \, (\ell \cos\theta - d) \cos\phi$$

**Initial value.** At $t = 0$: $\tfrac{1}{2} m \|\mathbf{v}_0^P\|^2 = \tfrac{1}{2} m \ell g$, $w(0) = \ell \cos(\pi/3) - d = \ell/2 - d$ and $\cos\phi(0) = 1/2$:

$$E_0 = \tfrac{1}{2} m \ell g - m g \left( \frac{\ell}{2} - d \right) \frac{1}{2} = m g \left( \frac{\ell}{2} - \frac{\ell}{4} + \frac{d}{2} \right) = \frac{m g \, (\ell + 2d)}{4}$$

**Conservation.** By the work-energy theorem, $\frac{d}{dt} \tfrac{1}{2} m \|\mathbf{v}\|^2 = \mathbf{W} \cdot \mathbf{v} + \mathbf{T} \cdot \mathbf{v}$. The tension term is zero by (f), and $\mathbf{W} \cdot \mathbf{v} = -\frac{d}{dt}(m g z_P)$ because gravity is conservative. Hence $E$ is **conserved**. Direct check, differentiating $E$ and inserting the equations of (e):

$$\frac{\dot{E}}{m} = \ell^2 \dot{\theta} \ddot{\theta} + w^2 \dot{\phi} \ddot{\phi} + w \dot{w} \dot{\phi}^2 - g \dot{w} \cos\phi + g w \sin\phi \, \dot{\phi}$$

with $\ell^2 \dot{\theta} \ddot{\theta} = -\ell \sin\theta \, \dot{\theta} \left( w \dot{\phi}^2 + g \cos\phi \right)$, $w^2 \dot{\phi} \ddot{\phi} = w \dot{\phi} \left( 2 \ell \sin\theta \, \dot{\theta} \dot{\phi} - g \sin\phi \right)$ and $\dot{w} = -\ell \sin\theta \, \dot{\theta}$. The terms in $\ell w \sin\theta \, \dot{\theta} \dot{\phi}^2$ add to $-1 + 2 - 1 = 0$, the terms in $g \ell \sin\theta \, \dot{\theta} \cos\phi$ add to $-1 + 1 = 0$, and the terms in $g w \dot{\phi} \sin\phi$ add to $-1 + 1 = 0$, so $\dot{E} = 0$.

**Angular momentum about $O$ is not conserved.** The torque of the tension about $O$ has the $\mathbf{k}_0$-component $-T \, (\mathbf{r}_0^Q \times \hat{\mathbf{u}}) \cdot \mathbf{k}_0 = -T d \sin\theta \, (1 - \sin\phi)$, which is non-zero at $t = 0$, and the weight has a non-zero horizontal torque. No component of $\mathbf{H}_O$ is a first integral in general, so the energy is the conserved quantity asked about.

PHASE 4

### Physical Interpretation & Verification

- **Numerical check** (Python, RK4 with step $10^{-4}$ s; $\ell = 2$ m, $d = 0.4$ m, $g = 9.81$ m/s$^2$, $\theta(0) = \phi(0) = \pi/3$, $\dot\phi(0) = 0$, hence $\dot\theta(0) = \sqrt{g/\ell} = 2.2147$ rad/s):

| Quantity | Result |
|---|---|
| Initial energy per unit mass | $E_0/m = 6.8670$ J/kg, equal to $g(\ell + 2d)/4 = 6.8670$ J/kg |
| Energy drift over 4 s | $\max \lvert E - E_0\rvert/m = 6.5 \times 10^{-11}$ J/kg |
| Tension and hanging length | $\min T/m = 2.94$ m/s$^2 > 0$ and $\min \rho = 0.38$ m $> 0$ |
| String-length constraint | $\lVert\mathbf{OQ}\rVert + \lVert\mathbf{QP}\rVert - \ell = 4 \times 10^{-16}$ m |
| Velocity formula of (c) | finite-difference velocity of $\mathbf{r}_0^P$ equals the closed form to four decimals |
| Angular momentum about $O$ | components of $\mathbf{H}_O/m$ vary (the $\mathbf{i}_0$ component runs from $-5.2$ to $7.3$ m$^2$/s) |

- **Limit $d \to 0$:** $Q \to O$, $w = \ell\cos\theta$ and $E = \tfrac{1}{2}m\ell^2(\dot\theta^2 + \cos^2\theta\,\dot\phi^2) - mg\ell\cos\theta\cos\phi$, the energy of a spherical pendulum of length $\ell$ in these angular coordinates.
- **Geometry:** $\theta = 0$ puts $Q$ at the foot of the perpendicular from $O$ (distance $d$), where the hanging length $\ell - d$ is maximal. The equations are singular when $w = 0$ (zero hanging length), which the hypothesis of a taut string over the edge excludes.
- **Dimensions:** $[\ell\ddot\theta] = [g] = \mathrm{m/s^2}$, $[T] = \mathrm{kg\cdot m/s^2} = \mathrm{N}$, $[E_0] = \mathrm{kg\cdot m^2/s^2} = \mathrm{J}$.

---

*Source: Problems.pdf (PDF page 27); key (part f only) on PDF page 76*